

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 6: Utility for Money

Notes on the Chapter

Chapter 6 concerns the most common applications of the expected-utility models of Chapter 5, to situations where the prizes or outcomes are amounts of money.

- It begins by developing the usual properties of utility functions for money: monotonicity, continuity, and concavity (risk aversion), and then develops the Arrow-Pratt measure of (absolute) risk aversion. (I do not cover relative risk aversion.) The first section closes by giving complementary results on first- and second-order stochastic dominance. (No proofs are provided in the text for this last part, but a very extended discussion in the *Student's Guide* fills in this gap.)
- The second section concerns induced utility for lotteries on money, induced from preferences over (lotteries on) consumption bundles. (Problems given at the end introduce the considerable difficulties encountered with discussing lotteries over prices.)
- A final section gives the most basic results on demand for insurance by a risk averse consumer; namely: The consumer will want full insurance if the insurance policy is actuarially fair, and less than full insurance if the policy is actuarially unfair. (Corresponding results for the demand for a single risky asset are given as a problem.)

Virtually nothing covered here is used in the remainder of this volume, although (of course) material from the first section is bound to become important in Volume II, if and when Volume II appears. Also, the discussion in Section 2 of this chapter foreshadows the general discussion of dynamic choice in Chapter 7 and might be useful to discuss, at least, so that students have a concrete example when they get to Chapter 7.

But, for the most part, you can pick and choose what you want to cover from this chapter, and/or you can supplement the material here with other topics of your own choosing.

Because of the potpourri nature of this chapter, I went back and forth on what to include, what to cover only in the *Guide*, and what to omit entirely. I can't say that I'm convinced I got it right, and feedback on the selection of topics would be welcome.

Notes on the Problems

This is clearly a "problem-rich" subject, and while I've tried to provide a reasonable selection of problems, I left out many interesting topics and directions. I call to your attention (in particular) Problems 6.5 and 6.6, where I ask the student to develop (within the context of simple random variables) the full theories of first- and second-order stochastic dominance. "Solutions" are provided in the *Student's Guide*; the scare-quotes here are because (1) students will need enough sophistication to understand language about random variables jointly realized on a common probability space, (2) students will need to know about the law of iterated conditional expectations and, most importantly, (3) when it comes to the one hard step—proving that if π is second-order stochastically dominant over ρ , then random variables with distributions π and ρ can be "realized" where the ρ -random variable is the π -random variable plus nonpositive conditional-mean noise—I sketch a direct construction for an example, but prove nothing.

Problem 6.7 makes the point that even if "underlying" preferences have an expected-utility representation, preferences induced from those underlying preferences may not do so, if decisions need to be taken before the uncertainty resolves. (Specifically, if you have to decide how much of one consumption good to purchase before you know how much money in total you'll have to spend, your preferences over lotteries on income will not have an expected-utility representation, even if your preferences over consumption lotteries do.) This point could be made in Chapter 5 and, in fact, it was very briefly mentioned there, as part of the discussion of temporal resolution of uncertainty. And it could be left to Chapter 7, since it involves dynamic choice to some extent. But since I discussed induced preferences for income here and made a big deal in the text out of the assumption that all the income-uncertainty resolves before consumption choices must be made, it seemed to fit best here. Of course, I'm prejudiced (since, long ago, I wrote about this point), but I think it is an important point, so I recommend this problem. (Added later: Indeed, this year I have an exceptionally able group of students, so I added some more pieces to this problem. See the discussion of the problem following if you are interested.)

Problems 6.13 and 6.14 lead the student from the simple ideas of this chapter in the directions of "gains from trade through securitization" and the (incorrect) notion that the law of large numbers is what makes insurance companies "work." But to undertake a full analysis of 6.14 requires that the student know at least about the theory of random walks and, in the end, the Martingale-Convergence Theorem. The application of the Martingale-Convergence Theorem is lovely, but it is probably beyond the capabilities of most students.

Solutions to Non-Starred Problems

- 6.2. The key here is that the consumer has constant absolute risk aversion; that is, his utility function takes the form $-e^{-\lambda x}$ for a single constant λ . Because we have one

certainly equivalent for him, we can find λ , which is a solution to the equation

$$-e^{-\lambda 2400} = \frac{1}{2}(-e^{-\lambda 5000}) + \frac{1}{2}(-e^{-\lambda 0}) = \frac{1}{2}(-e^{-\lambda 5000}) + \frac{1}{2}(-1).$$

We can't solve this equation analytically, but with Excel, we can easily find the solution numerically. I searched by hand (I tried Goalseek first, but it worked poorly), finding that $\lambda = 0.000032035$ worked pretty well. (The left-hand side of the equation above is -0.925997263 while the right-hand side was -0.92599734 .) I then used this value of λ to evaluate expected utilities and then certainty equivalents of the three gambles a, b, and c, finding that the CE of gamble a is \$2000 (of course), \$2460.03 for gamble b, and \$2220.42 for gamble c. This consumer will choose gamble b.

■ 6.4. For this function U , $U'(x) = x^{\alpha-1}$ and $U''(x) = (\alpha - 1)x^{\alpha-2}$, so the coefficient of absolute risk aversion is

$$-\frac{U''(x)}{U'(x)} = \frac{1 - \alpha}{x},$$

while the coefficient of relative risk aversion is x times this, or $1 - \alpha$. For this reason, this class of utility functions is called the constant relative risk aversion, or CRRA, utility functions. (The case $\alpha = 1$ is, of course, the risk neutral utility function, while $\alpha = 0$ "is" the case $-U''/U' = 1/x$, which one integrates to get $\ln(U'(x)) = -\ln(x) = \ln(1/x)$, hence $U'(x) = 1/x$ and $U(x) = \ln(x)$. (I'm ignoring the two constants of integration; with them, you'd have $U(x) = A \ln(x) + B$.)

■ 6.7. First, let me give the solution to the problem as posed in the text:

When the consumer's income is either 98 or 178, each with probability $1/2$, the optimal level of x_1 is 59; you can find this by setting the derivative in x_1 to zero (where $x_2 = y - x_1$) and then solving the resulting quadratic equation, or you can resort to numerical maximization. Either way, the expected utility is 8.33253016.

When the consumer's income is 126.948391819, the optimal level of x_1 is precisely half of that, or 63.4741959045. This gives an (expected) utility of 8.33253016, same as the first case.

And when the consumer's income is one of these three levels, each with probability $1/3$, the optimal level of x_1 is 60.2389939, for an expected utility of 8.33136843, less than the first two cases.

If the consumer's preferences over income lotteries had an expected-utility representation, her expected utility in this third case would have to be a convex combination of the first two levels of expected utility (and, since they are equal, it would therefore have to equal the two of them), because the third lottery is a convex combination of the first two. Hence, we conclude, the consumer does not have expected-utility preferences over income lotteries.

We can logically conclude that this is so, once we know three things: (a) Her expected utility in the first two cases are the same. (b) Her optimal choice of x_1 in the first two cases are different. (c) Her expected utility, given a specific lottery in y , is strictly concave in x_1 , hence the optimal choice of x_1 for a specific lottery is unique. (And we know all these things, for this problem.) Once we know a, b, and c, then when we mix the first two lotteries, no choice of x_1 can do as well as the optimal x_1 for the first lottery in the second case and vice versa. So whatever x_1 she picks, it leaves her worse off in one "piece" of the mixture or the other, or both. Her expected utility, therefore, is less than in the first two cases.

The problem set-up does say she has expected-utility preferences...over lotteries of consumption bundles. That's a key point. Someone can have expected-utility preferences over one set of lotteries (on one set of prizes) but not on another set of lotteries (on a different space of prizes). And, when uncertainty resolves after decisions must be taken, it is "rare" that the resulting induced preferences will be of the expected-utility variety. (For a description of when this rare thing happens, see Kreps and Porteus, JET, 1979.)

Now for an extension. Suppose that, instead of the utility function given in the problem, the utility function for the consumer is

$$U(x_1, x_2) = -e^{-0.0001x_1} - e^{-0.0001x_2}.$$

For this utility function, *do not* restrict x_1 and x_2 to be nonnegative. Suppose that (a) income is either 98 or 178, each with probability 1/2, or (b) income is a sure-thing 137.2, or (c) income is 98, or 178, or 137.2, each with probability 1/3. How does this compare with the situation in the problem as posed in the book? For this utility function on consumption, do the consumer's induced preferences in income lotteries, where she must choose x_1 before the uncertainty about y resolves, satisfy the mixture-space axioms?

As I said in the earlier discussion of problems in general, I included this extended version of the problem because the students in my class this year (2012) have so far been extremely able. The answer, broadly, is that this is one of the special utility functions where the induced preferences do have an expected utility representation. Seeing this, however, is difficult; my students this year saw that the "problem" they found with the log function utility went away in the numerical example, but they didn't know where to go with this observation in general. In the next several paragraphs, I'll provide the proof that these induced preferences do indeed have an expected utility representation. But it may be a bit much to expect most students to get there. (If you have one or two students who are outstanding, you might challenge them with this extension. And then you might challenge them further to tell you *all* the utility functions for which this is true.) Here is the solution for the extended problem:

If the consumer's income is either 98 or 178, each with probability 1/2, and if she chooses to consume x_1 in the first period, then her expected utility (of consumption) is

$$\frac{-e^{-0.001x_1} - e^{-0.001(98-x_1)}}{2} + \frac{-e^{-0.001x_1} - e^{-0.001(178-x_1)}}{2}.$$

I solved this numerically, and got an optimal $x_1 = 68.6$, for an expected utility of -1.8674 . And if $y = 137.2$ with probability 1, it is optimal to split the income evenly over the two periods, for $c_1 = 68.6$ (the same as before!) and a utility level of -1.8674 (the same as before!). So it is clear (why?) that if the income level is 98, or 178, or 137.2, each with probability 1/3, the optimal thing to do is to set $c_1 = 68.6$, giving expected utility -1.8674 . (And, if you don't believe me, do this numerically. But you should believe me.)

So what's going on here? For the utility function in the problem (the utility function with logs), we had two income lotteries which gave the same expected utility but had different levels of c_1 . So when we took a mixture of the two lotteries, the expected utility had to fall. Here, two income lotteries that give the same level of expected utility have the same value of c_1 , so we can use that level of c_1 for any convex combination of the two and get the same expected utility.

Is this general? It is. Let y denote the random (variable) income. Then if the decision maker chooses to consume c_1 , her expected utility (with this utility function) is

$$-e^{-0.001c_1} + \mathbf{E}[-e^{-0.001(y-c_1)}] = -e^{-0.001c_1} + e^{0.001c_1} \mathbf{E}[-e^{-0.001y}],$$

where $\mathbf{E}$ stands for expectation over the random variable y . Please examine that last step carefully: Because of properties of the exponential function, we get this very nice form. This has the form

$$-e^{-0.001c_1} + Ke^{0.001c_1},$$

where K is the constant $\mathbf{E}[-e^{-0.001y}]$. The first-order condition on this (when you maximize in the choice of c_1) is

$$0.001e^{-0.001c_1} + 0.001Ke^{0.001c_1} = 0, \quad \text{or} \quad Ke^{0.001c_1} = -e^{-0.001c_1} \quad \text{or} \quad -e^{-0.002c_1} = K,$$

which has solution

$$c_1 = \ln(-K)/(-0.002),$$

and for which the value of the objective function at the solution is

$$\begin{aligned} -e^{-0.001\ln(-K)/(-0.002)} + Ke^{0.001\ln(-K)/(-0.002)} &= -e^{\ln(-K)/2} + Ke^{-\ln(-K)/2} \\ &= -(-K)^{1/2} + K \cdot (-K)^{-1/2} = -2(-K)^{1/2}, \end{aligned}$$

where, you will recall, $K = -\mathbf{E}[e^{-0.001y}]$. The value of the objective function, then, is a monotonic (increasing) function in K , and two income lotteries, call them y and y' , give the same level of (induced) expected utility if and only if they give the same level of K . But if they give the same level of K , they induce the same level of c_1 , just as we saw in the example.

More than that, suppose y and y' are two income lotteries. Write $K(y)$ for $\mathbf{E}[-e^{-0.001y}]$. Note that

$$\begin{aligned} K(y) \geq K(y') \quad \text{iff} \quad -K(y') \geq -K(y) \\ \text{iff} \quad (-K(y'))^{1/2} \geq (-K(y))^{1/2} \quad \text{iff} \quad -2(-K(y))^{1/2} \geq -2(-K(y'))^{1/2}. \end{aligned}$$

That is, the (induced) expected utility of the consumer if her income lottery is y is at least as large as her (induced) expected utility under y' if and only if $K(y) \geq K(y')$. But $K(y)$ takes the form of expected utility, for a utility function defined on income lotteries. The consumer with the second utility function is an expected utility maximizer when it comes to lotteries on her income!

■ 6.9. *It depends.* If the consumer buys insurance level a , her expected utility is

$$v(a) = \pi V(y - a\delta) + (1 - \pi) \sum_{x \in \text{supp}(\Delta), x \neq 0} \rho(x) V(y - x - a\delta + aB),$$

where $\rho(x)$ is the conditional probability that the loss is x , conditional on there being a loss at all. This function v is concave and differentiable, so the first-order condition for a maximum is $v'(a) = 0$. So we evaluate $v'(a)$:

$$v'(a) = (B - \delta)(1 - \pi) \sum_{x \in \text{supp}(\Delta), x \neq 0} \rho(x) V'(y - x - a\delta + aB) - \delta\pi V'(y - a\delta)$$

Assuming the contract is actuarially fair, or $(1 - \pi)B = \delta$, we have $(B - \delta)(1 - \pi) = (1 - \pi)B - (1 - \pi)\delta = \delta - (1 - \pi)\delta = \pi\delta$. so the $v'(a)$ simplifies to

$$v'(a) = \delta\pi \left[\sum_{x \in \text{supp}(\Delta), x \neq 0} \rho(x) V'(y - x - a\delta + aB) - V'(y - a\delta) \right].$$

At $a = 1$, we have

$$v'(1) = \delta\pi \left[\sum_{x \in \text{supp}(\Delta), x \neq 0} \rho(x) V'(y - x - \delta + B) - V'(y - \delta) \right].$$

The parameter B is entirely free here. If B is close to zero, then since all x in the support of Δ that are not zero are positive, we would have $v'(1) > 0$, and the consumer wants to "overinsure." If B is larger than every x in the support of Δ , we find $v'(1) < 0$, and the consumer will choose an $a < 1$.

■ 6.11. (a) The problem here is to prove that a solution exists; its uniqueness (once we know there is a solution) is guaranteed by Proposition 6.18(f). To show that there is a solution, write out the derivative of $v(a)$ as defined in equation (6.3):

$$v'(a) = \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)e^{-\lambda[a(\theta-r)+rw]}\pi(\theta).$$

Note that this is a continuous function of a , and that $v'(0) = \sum_{\theta} \lambda(\theta - r)e^{-\lambda rw}\pi(\theta) = \lambda e^{-\lambda rw} \sum_{\theta} (\theta - r)\pi(\theta) > 0$, since we are assuming that the expected value of the return on the risky asset exceeds r . We will know there is a finite solution by the Intermediate-Value Theorem, then, if we show that $v'(a)$ is eventually negative. So consider the terms in the sum. For $\theta > r$, as a rises, the term $e^{-\lambda[a(\theta-r)+rw]}$ vanishes. And for $\theta < r$, this term goes to infinity. Eventually (there are only finitely many of these), the terms for $\theta < r$ will dominate the sum, giving a negative value.

(b) The first-order condition that describes the solution is

$$\begin{aligned} 0 &= \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)e^{-\lambda[a(\theta-r)+rw]}\pi(\theta) = \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)e^{-\lambda[a(\theta-r)]}e^{-\lambda rw}\pi(\theta) \\ &= e^{-\lambda rw} \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)e^{-\lambda[a(\theta-r)]}\pi(\theta). \end{aligned}$$

But the last term equals zero if and only if $0 = \sum_{\theta} \lambda(\theta - r)e^{-\lambda[a(\theta-r)]}\pi(\theta)$, which is entirely independent of the value of w .

(c) Use the notation $v'(a; \lambda)$ for the summation on the right-hand side of the display in part (a); that is, $v'(a; \lambda)$ is the derivative of $v(a)$ (in a), treating λ as a parameter. We know that the solution to the problem for a given λ is where $v'(a; \lambda) = 0$. We also know that $v'(a; \lambda)$ is a nonincreasing function in a for each λ , since $v(a; \lambda)$ is concave. With all this, we will demonstrate that

$$\begin{aligned} \frac{\partial v'(a; \lambda)}{\partial \lambda} &= \sum_{\theta \in \text{supp}(\pi)} (\theta - r)e^{-\lambda[a(\theta-r)+rw]}\pi(\theta) \\ &\quad - \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)[a(\theta - r) + rw]e^{-\lambda[a(\theta-r)+rw]}\pi(\theta) \end{aligned}$$

is strictly less than zero where $v'(a; \lambda) = 0$. This shows that an increase in λ pushes v' negative, so the solution to $v'(a; \lambda) = 0$ is a nonincreasing function of λ . (To be very

formal about this requires the Implicit-Function Theorem, which we will get to later in the book. Since this is a one-dimensional problem (that is, we are eliminating a one-dimensional variable), and since we know that $v'(a; \lambda)$ is nonincreasing in a , this degree of formality is unnecessary. But for the interested reader, Appendix 7 outlines the Implicit-Function Theorem.)

So, to show that the partial derivative is strictly negative, note that the first summation will be zero where $v'(a; \lambda) = 0$, so it can be ignored. And in the second summation, break this into two pieces, namely

$$\sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)[a(\theta - r)]e^{-\lambda[a(\theta - r) + rw]}\pi(\theta) + \sum_{\theta \in \text{supp}(\pi)} \lambda(\theta - r)[rw]e^{-\lambda[a(\theta - r) + rw]}\pi(\theta).$$

The second of these is zero, as well, where $v'(a, \lambda) = 0$. So we have

$$\frac{\partial v'(a; \lambda)}{\partial \lambda} = - \sum_{\theta \in \text{supp}(\pi)} \lambda a(\theta - r)^2 e^{-\lambda[a(\theta - r) + rw]}\pi(\theta),$$

which is obviously negative.

■ 6.13. (a) Use Taylor's Theorem to write $U(w + \alpha(x - \beta E)) = U(w) + U'(w)(\alpha(x - \beta E)) + o(\alpha(x - \beta E))$. Therefore, if the consumer purchases the small share, her expected utility will be

$$\begin{aligned} \sum_{x \in \text{supp}(\pi)} \left[\pi(x) [U(w) + U'(w)(\alpha(x - \beta E)) + o(\alpha(x - \beta E))] \right] \\ = U(w) + \alpha U'(w) \left[\sum_x (x - \beta E) \pi(x) \right] + o(\alpha). \end{aligned}$$

(The sum of the terms $o(\alpha(x - \beta E))\pi(x)$ is $o(\alpha)$ because $k o(k'\alpha)$ is $o(\alpha)$ for all constants k and k' and the sum of finitely many terms that are $o(\alpha)$ is $o(\alpha)$.)

But the summation inside the square brackets is $(1 - \beta)E$, and as $\beta < 1$, this term is strictly positive. So the consumer's expected utility if she purchases the small share exceeds $U(w)$, her utility if she does not, by $\alpha U'(w)(1 - \beta)E + o(\alpha)$, which is greater than zero for all small-enough α .

(b) Both nondifferentiability of U at w and a random income for the consumer can upset this result. If U is not differentiable, we would need to look at Taylor's Theorem in terms of left- and right-hand derivatives at U (recall that U is concave, so it will have left- and right-hand derivatives). The derivative to the left (for x that are less than βE) will be larger than the derivative to the right (which applies for x greater than βE), so there is no guarantee that the coefficient on α will be positive; we weight the losses relative to E more than the gains.

And if the consumer has random income: If that income is independent of the prizes given by π , the result will go through; I leave it to you to do the math. But if the income is not independent, again we will be unable to conclude that the coefficient on α is positive. In fact (and a bit loosely), if the consumer's income is positively correlated with the results of the lottery π , then the "weight" on terms $x - \beta E$ for larger values of x will be less than the weight for smaller values of x , which will diminish the value of the small share of π . If, on the other hand, the correlation is negative, this will enhance the value of the small share. (Examples are easy to construct.)